

AREMU OLUWATOBILOBA

AI Engineer | Backend Engineer

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PROFESSIONAL SUMMARY

AI Engineer with a strong backend foundation, specialising in building intelligent systems powered by Large Language Models. Experienced designing and shipping end-to-end AI pipelines, production-style REST APIs with FastAPI, database-backed services, and LLM-integrated endpoints with schema validation, retry logic, and cost controls. Comfortable across the full stack, from authentication and containerised deployment to agent design and tool-connected AI systems. Motivated, fast-learning, and driven by practical impact through software engineering.

TECHNICAL SKILLS

AI & Machine Learning: RAG Pipelines, LLM Integration, Prompt Engineering, Semantic Search, Vector Databases (ChromaDB), Sentence Transformers, Groq API, Ollama (LLaMA 3.1), Agentic AI Systems, Model Context Protocol (MCP)

Backend Development: Python, FastAPI, Flask, REST API Design, PostgreSQL, SQLite, MongoDB, Uvicorn

Authentication & Security: Supabase Auth, JWT, bcrypt password hashing

DevOps & Deployment: Docker, Docker Compose, Render, Vercel, Hugging Face Spaces, Git, GitHub

Frontend & Interfaces: HTML/CSS/JavaScript, Gradio

EXPERIENCE

Backend AI Engineering Intern — FlyRank AI (Remote) 2026 – Present

- Built and deployed a full CRUD REST API in Python (FastAPI) with request validation, then migrated storage from in-memory to a real SQLite database with zero change to API behaviour; documented and published to GitHub.
- Built a polite, production-style web scraping pipeline (Requests, BeautifulSoup, Pydantic) with a fetch, extract, normalise, validate, store, and report flow; handled rate-limiting and page failures without crashing the run.
- Implemented secure authentication with Supabase Auth and JWTs, covering signup, login, logout, and protected routes, verified via Swagger UI.
- Designed and built a personal AI agent (a Weekly Review Assistant) pulling data through a live connector, following a written design spec with eval cases and safety guardrails.
- Integrated an LLM into a backend endpoint with a parse-validate-repair pipeline, timeout and backoff/jitter retry logic, structured cost logging, and a kill switch; verified against a written evaluation set.
- Containerised a FastAPI + PostgreSQL backend with Docker Compose, connecting the service to Postgres through a repository layer with zero change to the API layer, and confirmed data persistence across a restart.
- Studied the workflow-vs-agent distinction and MCP's core primitives; connected a live MCP connector and ran real tool-use tasks beyond plain chat.

AI Engineering Facilitator — LearnItDigitalHub (Remote) 2026 – Present

- Facilitate training in AI Engineering & Intelligent Systems, and in Deep Learning & Advanced Generative AI.
- Deliver the AI Engineering track through structured video lessons and hands-on assignments covering prompt engineering, LLMs, RAG, and vector databases.
- Responsible for expanding course curriculum with supplementary material to support student learning outcomes.

PROJECTS

MyScholar AI — RAG Pipeline for Academic Document Q&A 2026

<https://github.com/Aremuoluwatobi/rag-pipeline> • <https://huggingface.co/spaces/AremuTaiwo/Rag-Pipeline>

- Built a full end-to-end Retrieval-Augmented Generation system enabling students and researchers to query academic PDF and DOCX documents conversationally.
- Designed an ingestion pipeline covering text extraction, chunking, embedding generation, and vector storage using ChromaDB and Sentence Transformers.

- Integrated Groq API (llama-3.1-8b-instant) for context-aware response generation grounded strictly in document content, with exponential backoff retry logic for rate limits.
- Deployed on Hugging Face Spaces using Docker, with documents stored and retrieved from a Hugging Face Dataset at runtime; built a FastAPI backend for local testing and programmatic access.

Axiomat AI — AI Assistant Embedded in Personal Portfolio 2026

<https://aremu-taiwo-portfolio.vercel.app/>

- Built a full-stack personal portfolio site with an embedded AI chat assistant that answers visitor questions about the author's background and projects.
- Designed a two-tier response system: instant canned replies for common messages, LLM fallback grounded in a dedicated knowledge base document to prevent hallucination, plus a safety fallback if the model call fails.
- Built the backend with FastAPI and deployed it on Render; built the frontend with vanilla HTML/CSS/JavaScript and deployed it on Vercel, connected via a REST API call.

Inventory Management API — Backend + AI Categorisation System 2026

<https://github.com/Aremuoluwatobi/inventory-api>

- Built a Python backend API for managing inventory operations with full CRUD functionality using Flask and PostgreSQL.
- Integrated LLaMA 3.1 via Ollama to automatically categorise inventory items into predefined categories, using prompt engineering for accurate, structured AI-driven classification.
- Implemented JWT-based authentication and bcrypt password hashing for secure admin access.

CERTIFICATIONS

20+ courses completed through the FlyRank AI Backend Engineering track and Anthropic Academy, including the certified “AI Fluency: Framework & Foundations” course. Full list with verification links available on the portfolio site.

EDUCATION

Bachelor of Arts in History — University of Ilorin, 2023 – Present

INTERESTS

- AI Engineering and Large Language Model Applications
- Backend Engineering and Scalable API Systems
- Applying AI to solve real-world problems across different fields and industries
- Emerging AI — agentic workflows, multi-agent systems, and autonomous pipelines